Are LLMs Good Zero-Shot Fallacy Classifiers?

Fengjun Pan[1,2]* **Xiaobao Wu**[2]* **Zongrui Li**[1] **Anh Tuan Luu**[2]✉
[1]Interdisciplinary Graduate Programme, Nanyang Technological University, Singapore
[2]College of Computing and Data Science, Nanyang Technological University, Singapore
panf0004@e.ntu.edu.sg xiaobao002@e.ntu.edu.sg
zongrui001@e.ntu.edu.sg anhtuan.luu@ntu.edu.sg

arXiv:2410.15050v1 [cs.CL] 19 Oct 2024

Abstract

Fallacies are defective arguments with faulty reasoning. Detecting and classifying them is a crucial NLP task to prevent misinformation, manipulative claims, and biased decisions. However, existing fallacy classifiers are limited by the requirement for sufficient labeled data for training, which hinders their out-of-distribution (OOD) generalization abilities. In this paper, we focus on leveraging Large Language Models (LLMs) for zero-shot fallacy classification. To elicit fallacy-related knowledge and reasoning abilities of LLMs, we propose diverse single-round and multi-round prompting schemes, applying different task-specific instructions such as extraction, summarization, and Chain-of-Thought reasoning. With comprehensive experiments on benchmark datasets, we suggest that LLMs could be potential zero-shot fallacy classifiers. In general, LLMs under single-round prompting schemes have achieved acceptable zero-shot performances compared to the best full-shot baselines and can outperform them in all OOD inference scenarios and some open-domain tasks. Our novel multi-round prompting schemes can effectively bring about more improvements, especially for small LLMs. Our analysis further underlines the future research on zero-shot fallacy classification. Codes and data are available at: https://github.com/panFJCharlotte98/Fallacy_Detection.

1 Introduction

A fallacy is a defective argument derived from erroneous or invalid reasoning that may appear to be reasonable but are, in fact, logically unsound or faulty (Woods, 2004; Damer, 2008; Van Eemeren et al., 2009; Hamblin, 2022). Figure 1 illustrates examples of different fallacy types, for instance, a fallacy type of Circular Reasoning: *I am a great leader because I make great leadership decisions.*

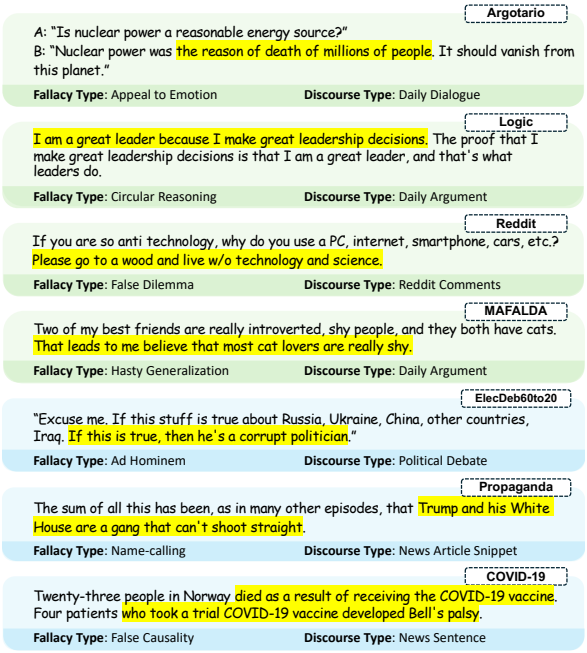

Figure 1: Examples of fallacies and their types from existing datasets.

Fallacies commonly appear in various scenarios, such as news articles (Musi and Reed, 2022), advertisements (Danciu et al., 2014), propaganda (Walton, 1997), politics (Blassnig et al., 2019), and social media (Hidayat et al., 2020). They could be intentionally exploited to disseminate misinformation (Musi et al., 2022a), manipulate public opinions, undermine rational discussions, and influence critical decision-making (Visser et al., 2020). In consequence, detecting and classifying fallacies becomes an imperative challenge.

However, existing fallacy classification methods typically follow a full-shot supervised fine-tuning manner, limited by three issues: (**i**) They require sufficient labeled data for training (Goffredo et al., 2022; Alhindi et al., 2023a), but collecting these data is often time-consuming and expensive. This is because annotating fallacy data typically rely on expert knowledge, due to the complex and esoteric nature of fallacies (Helwe et al., 2023). (**ii**) They cannot well generalize to out-of-distribution (OOD)

*Equal contribution. ✉ Corresponding Author.

fallacies and discourse types (Liu et al., 2021) due to the inherent limitation of supervised learning. Once given an unseen fallacy class or discourse from other domains, they have to retrain a new model from scratch, which costs considerable computational and time resources. (**iii**) They are susceptible to imbalanced data (Alhindi et al., 2023b). They could reach high performance on the dominant fallacy types while low on infrequent ones. As a result, full-shot fallacy classifiers cannot fulfill real-world application scenarios.

Motivated by the above issues, in this paper, we explore zero-shot fallacy classification, *i.e.*, classifies fallacies without training data. Specifically, we concentrate on Large Language Models (LLMs) as they have been extensively pretrained and possess wide knowledge and strong reasoning abilities. To investigate LLMs' performance on this task, we consider two kinds of prompting schemes. First, we employ a **single-round prompting scheme**. We simply prompt LLMs to classify fallacies either with or without manually crafted fallacy type definitions.

Second, to elicit the inherent fallacy-related knowledge and reasoning abilities of LLMs, we further propose diverse **multi-round prompting schemes**: we instruct LLMs to analyze and classify fallacies through definition generation, general fallacy analysis (with warm-up), premise & conclusion analysis, and Chain-of-Thought. Under these prompting schemes, we observe that zero-shot prompted LLMs can outperform or achieve comparable performances with SOTA full-shot fine-tuned T5 baselines on some open-domain benchmark datasets and can generally achieve sub-optimal performances on hard domain-specific datasets. More importantly, LLMs consistently demonstrate advantage over full-shot trained baselines on OOD inferences on low-resource fallacy classification tasks. Besides, our novel multi-round prompting schemes can effectively improve LLMs' classification performance compared to the single-round prompts, especially for small LLMs, *e.g.*, Llama3, Qwen2.5 and Mistral. We conclude our main contributions as below:

- We propose diverse novel prompting schemes, including both the basic single-round and the advanced multi-round ones, that are sufficiently effective in eliciting the fallacy-related knowledge and reasoning abilities of LLMs for zero-shot fallacy classification.
- We are the first to conduct extensive experiments with representative LLMs on existing fallacy benchmark datasets and provide overviews and insights concerning the boundary of LLMs' zero-shot fallacy classification performances. We show that we can resort to LLMs as potential zero-shot fallacy classifiers that solely rely on LLMs' inherent knowledge without computationally intensive training full-shot models from scratch.
- We provide detailed analysis on LLMs, datasets, and prompting schemes and summarize a general guidance for choosing the potentially most effective prompting schemes for different fallacy classification scenarios with respect to data domains and LLM types, inspiring future research on zero-shot fallacy classification and other relevant linguistic reasoning tasks.

2 Related Work

Fallacy Classification Fallacy detection and classification is an emerging natural language processing task that has received increasing attention. Many fallacy datasets across different discourse genres and various domains have been created in a line of work (Habernal et al., 2017, 2018b; Da San Martino et al., 2019; Sahai et al., 2021; Goffredo et al., 2022; Jin et al., 2022; Alhindi et al., 2023a; Musi et al., 2022b; Helwe et al., 2023). Early fallacy classification methods are mainly based on traditional machine learning models (Wu et al., 2020, 2022, 2023, 2024e,d,c,a,b), *e.g.*, SVM, Bi-LSTM (Habernal et al., 2018a) and CNN (Habernal et al., 2018b). Deep learning methods based on models with Transformers architecture *e.g.*, BERT (Devlin et al., 2018), RoBERTa (Liu et al., 2019), T5 (Raffel et al., 2020) have been proposed in recent works (Da San Martino et al., 2019; Sahai et al., 2021; Goffredo et al., 2023; Jin et al., 2022; Alhindi et al., 2023a). However, all these methods are trained in a supervised manner that is highly reliant on the availability and sufficiency of labeled fallacy data, rendering them struggling on OOD fallacy and discourse types. Recently, LLMs' abilities related to fallacy understanding and detection have been explored (Payandeh et al., 2023; Alhindi et al., 2023b; Lim and Perrault, 2024; Lalwani et al., 2024; Ruiz-Dolz and Lawrence, 2023) but not evaluated in-depth considering prompt techniques.

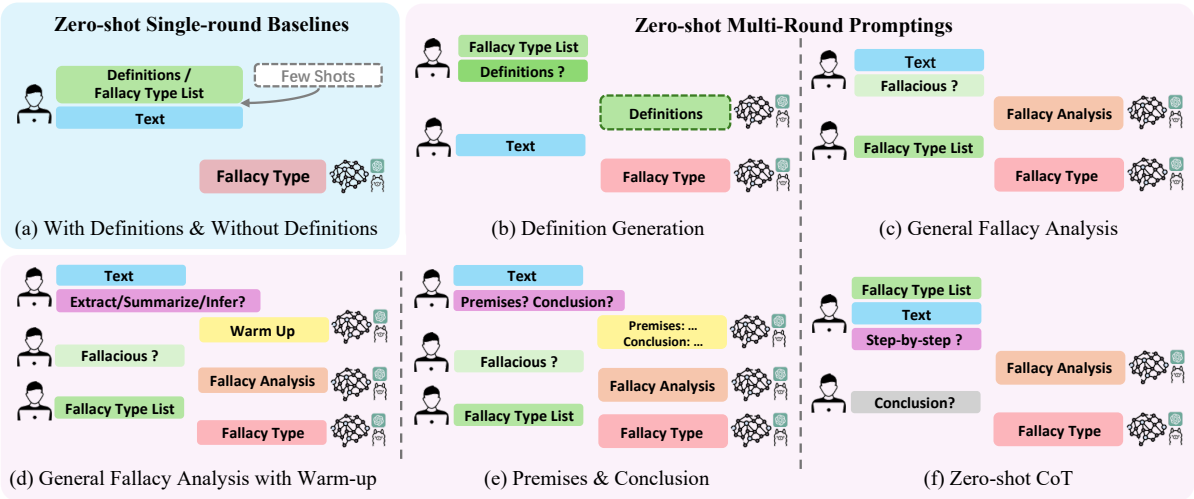

Figure 2: Illustration of single-round and multi-round prompting schemes. (**a**): Prompt LLMs to classify with or without fallacy type definitions. (**b**): Prompt LLMs to generate fallacy type definitions and then classify. (**c**): Prompt LLMs to analyze the input discourse and then classify. (**d**): Prompt LLMs to warm up (extract, summarize, or infer), analyze the input discourse, and then classify. (**e**): Prompt LLMs to extract the premises and conclusion, analyze the input discourse, and then classify. (**f**): Prompt LLMs to reason step by step to classify and then draw an answer.

Large Language Models The ability of LLMs in logical reasoning has been quantitatively or qualitatively evaluated, *e.g.*, commonsense causality reasoning (Kıcıman et al., 2023; Talmor et al., 2020; Willig et al., 2022) and abstract reasoning (Gendron et al., 2023; Pan et al., 2023; Wu et al., 2024f). Particularly, evidence (Bubeck et al., 2023) has verified the existence of inconsistency in its reasoning process (same for the latest LLMs like GPT-4). This motivates us to guide LLMs through the proposed multi-round prompting schemes. Our work distinguishes from the previous in two points: (**i**) instead of a general reasoning task, we focus on the fallacy classification, a more advanced reasoning task that requires not only expert knowledge of but also rigorous logical reasoning. (**ii**) Apart from the common single-round prompting, we propose the novel multi-round prompting schemes that bring along noticeable improvements for small LLMs.

3 Methodology

In this section, we apply LLMs for zero-shot fallacy classification by two kinds of prompting schemes: **Zero-shot Single-Round Prompting** and **Zero-shot Multi-Round Prompting**.

3.1 Zero-shot Single-Round Prompting

We first propose zero-shot single-round prompting schemes. Following the state-of-the-art fine-tuning baseline (Alhindi et al., 2023a), the single-round prompt consists of three components, as shown in Figure 2 (a): (i) A label space that defines a limited number of fallacy types; (ii) An input fallacious discourse to be classified; (iii) The task and output format instructions. For the label space, we consider two settings: (i) **Without Definitions**, which merely enumerates all the viable fallacy types as an option list, for example, *1. Appeal to Emotion, 2. Ad Hominem, 3. False Dilemma ....* (ii) **With Definitions**, which means that we provide the natural language definition of each fallacy type, for example, *1. Appeal to Emotion is a fallacy when someone attempts to argue or persuade by using emotive language to arouse non-rational sentiments within the intended audience. 2. Ad Hominem is a fallacy when someone attacks the others' characters or motives instead of addressing the substance of their arguments...* See the used single-round prompt templates in Table 29.

3.2 Zero-shot Multi-Round Prompting

Besides, we propose zero-shot multi-round prompting schemes. This is motivated by the complexity nature of the fallacy classification task as it involves reading comprehension, information extraction, logical reasoning, knowledge recall, and pattern recognition. Therefore, we introduce the following multi-round prompting schemes that aim to elicit LLMs' inherent knowledge and reasoning abilities on fallacy. Figure 2 summarizes all these schemes.

Definition Generation As illustrated in Figure 2 (b), we prompt LLMs to generate the definition for each fallacy given in the fallacy type list in the